Where does Vera shine? — comparison report v2

HQ Vera · AMD EPYC 9575F · AMD EPYC 9J45 (Turin) · AWS Graviton5, with the create-path fixes on

Data as of 2026-10-14 23:30 UTC, from the three working sessions (Vera + 9575F, Turin, Graviton5). Supersedes the 15:05 status report.

What changed since the last report. All four hosts now ran the **same Daytona build** (`f0611fa13` + bench patches) with the four create-path fixes switchable by flag (3260 init-disk no-fsync, 3261 cgroup pool, 3265 permanent neighbour entry, 3269 API listen backlog 4096). The 9575F was brought onto that build and measured fixes off vs on; Vera and the 9575F were run at 2,000 sandboxes; Turin got the fixes and was rerun at 1k/2k, and its density ceiling was pushed to 9,750 live sandboxes. Graviton5 was rerun fixes off vs on. **Green bold** = best value in a row. **Startup** = 1,000 (or 2,000) sandboxes created at once by 16 client processes (`mpfleet.py`), probe `echo ready`. **Idle image** (`alpine:3.20`) measures the platform; the **agent image** runs a ≈ 3–4 core-second job in every sandbox at boot, so it measures startup under CPU load.

1. Verdict

Question	Winner	Evidence (fixes on unless noted)
Throughput (jobs/s, dense agent ladder)	**Vera** (Turin not yet measured)	Vera **26.7** vs Graviton5 22.9 (+16 %) vs 9575F 8.0 (×3.3). Holds at 2,000 sandboxes: 26.1 vs 8.0. Fixes do not change throughput anywhere.
Per-core speed	**Vera**	0.15 jobs/s per physical core vs 0.13 (9575F) and 0.12 (Graviton5); fastest single job (5.9 s vs 6.6–6.7 s).
Platform startup, idle image, 1k	**9575F** / Graviton5	Typical create p50: Graviton5 **0.85 s**, 9575F **1.06 s**, Vera 1.57 s, Turin 2.35 s. All-ready: Graviton5 1.7–2.3 s, 9575F 2.1 s, Vera 2.7 s, Turin 3.9 s.
Platform startup, idle image, 2k	**9575F**	p50 / all-ready: 9575F 2.4 / 4.6 s, Vera 2.9 / 6.6–7.2 s, Turin 3.5 / 7.1–7.2 s.
Startup under load (agent image)	**Turin**	1k all-ready: Turin **7.9 s**, Graviton5 10–12 s, 9575F 19 s, Vera 22–28 s (one run 61 s). 2k: Turin **21 s**, Vera 50–55 s, 9575F 58–71 s.
Density (live sandboxes)	**Turin**	Turin held **8,000** cleanly and peaked at 9,750 — stopped by the runner's heartbeat, not hardware. Vera and the 9575F ran 2,000 with ≤ 1 failure per 16,000 jobs.
Reliability	9575F, Turin, Graviton5	0 failures in thousands of startups on x86 and Graviton5. Vera shows ≈ 0.2 % post-resume guest wedges (3213) that hold "all ready" open.

Bottom line. Vera shines on **sustained CPU-bound sandbox work**: the highest throughput measured so far and the fastest cores. It does **not** lead on platform startup — with the fixes on, the 9575F and Graviton5 start sandboxes faster — nor on density. The decisive gap is **Turin's throughput ladder**: Turin's agent-image bursts (a CPU-throughput proxy) are 2.5–3× faster than Vera's, so Turin may match or beat Vera on throughput. Until that ladder runs, "Vera has the highest throughput" holds only against Graviton5 and the EPYC 9575F / 9755.

2. Hosts and build

	HQ Vera	EPYC 9575F	Turin (OCI phoenix-e6)	Graviton5 (AWS)
CPU	NVIDIA Vera, 2 × 88 c, SMT2 (352 thr)	AMD Zen 5, 1 × 64 c, SMT2 (128 thr)	AMD EPYC 9J45, 2 × 96 c, SMT2 (384 thr)	Neoverse-V3, 1 × 192 c, no SMT
Memory	1,479 GiB	1,133 GiB	2,267 GiB	754 GiB
Host kernel / pages	6.17 nvidia / 64 KiB	7.0 / 4 KiB	6.17 oracle / 4 KiB	6.17 aws / 4 KiB (64 KiB A/B done)
Guest kernel	6.12.34-rlp1 arm64	6.8.12-rlp3 x86	6.8.12-rlp3 x86	6.12.34-rlp1 arm64
`RLP_VM_MAXACK`	256 (default)	**2,048**	256	256
Client HTTP pool / disk	1,024 / 1 GiB	1,024 / 1 GiB	512 / 64 MiB	1,024 / 1 GiB

All four: Daytona `f0611fa13` + EROFS-4k (1060) + `RLP_FC_SOCK_WAIT_MS=30000` (3212) + the four create-path fixes behind flags; Firecracker v1.16.1; daemon `rlp-rs-0.4.1`; `VM_CONCURRENCY=512`; balloon off; unpinned (whole box). Vera's and Graviton5's runner binaries are byte-identical. Turin's 3269 patch was implemented independently (same flag, same effect).

3. Startup

3.1. 1,000 sandboxes, idle image — fixes off → on (seconds; reps separated by "/")

Metric	HQ Vera	EPYC 9575F	Turin	Graviton5
create p50, off	2.07 / 2.39 / 2.32	2.47 / 2.58 / 2.49	3.72 / 3.68	1.46 – 2.18
create p50, on	1.51 / 1.62 / 1.59	1.10 / 0.98 / 1.09	2.37 / 2.34	**0.81 – 0.90**
create p95, off	3.30 / 3.76 / 3.46	3.36 / 3.91 / 3.76	7.71 / 7.65	3.11 – 3.55
create p95, on	2.54 / 2.61 / 2.57	1.59 / 1.60 / 1.57	3.76 / 3.76	**1.54 – 1.56**
all ready, off	4.70 / 3.86 / 3.66	3.82 / 4.00 / 4.53	7.95 / 7.96	3.57 / 3.60 / 3.63
all ready, on	10.42[a] / 3.88 / 2.69	2.26 / 2.12 / **2.05**	3.88 / 3.87	2.32 / **1.68** / 7.63[b]
p50 gain from fixes	−30 %	−57 %	−36 %	≈ −60 %
failures (on)	1 in 3,000	0	0	0

[a] held open by one wedged guest (3213/3229). [b] 5 sandboxes waited 5.1–5.7 s for their toolbox at one moment — a new tail seen only with fixes on, unexplained. Turin: all-ready 7.95 ⟶ 3.88 s (−51 %) closes the previous report's open item: its ≈ 8 s was measured without the fixes. Its remaining dominant phase is snapshot `load` (≈ 0.5 s p50).

3.2. 1,000 sandboxes, agent image (startup under load) — fixes on

Metric	HQ Vera	EPYC 9575F	Turin	Graviton5
create p50	13.72 / 11.84 / 7.55	6.60 / 6.63 / 8.28	**2.54 / 2.51**	5.43 – 5.49
create p95	23.90 / 26.05 / 21.73	16.61 / 18.50 / 16.53	**5.77 / 5.75**	7.48 – 11.61
all ready	61.02[a] / 28.11 / 21.82	18.85 / 19.14 / 18.77	**7.84 / 7.88**	10.11 / 10.79 / 12.31
all ready, fixes off	21.82 / 21.98 / 25.75	20.35 / 20.24 / 20.56	7.90 / 8.01 / 7.97	10.10 / 11.77 / 11.83

The fixes do not move agent-image numbers (≤ 3 %): the burst is bound by the in-guest boot job, i.e. by host CPU throughput.

3.3. 2,000 sandboxes at once — fixes on

Metric	HQ Vera	EPYC 9575F	Turin	Graviton5
idle: create p50	3.00 / 2.79	**2.46 / 2.34 / 2.40**	3.50 / 3.47	not run
idle: create p95	6.40 / 6.57	**3.63 / 3.57 / 3.57**	6.89 / 6.79	not run
idle: all ready	6.59 / 7.23 / (lost run)[c]	**4.62 / 4.56** / 17.65	7.22 / 7.13	not run
idle: all ready, fixes off	—	—	23.22	not run
agent: create p50	29.4 / 27.5	19.7 / 19.6	**7.00**	not run
agent: all ready	50.2 / 54.5	58.1 / 70.7	**21.41**	not run
failures	2 (toolbox dial timeouts) + 1 lost run	0	0	—

[c] one sandbox never answered; the client was killed after 20 min (wedged-guest pattern).

4. Throughput — dense agent ladder (jobs/s)

Published recipe: 45 jobs, 8 episodes per sandbox, hold-then-exec, 0.025 CPU (max 1), 64 MiB (max 4 GiB). Unpinned.

Host · arm	176	352	704	1,056	2,000	Note
HQ Vera · fixes on	15.17	**26.65**	25.62	—	**26.11**	2k: job p50 34 s, 1 failure in 16,000
HQ Vera · fixes off	15.0[d]	26.67	25.51	—	—	
EPYC 9575F · fixes on	8.01	8.05	8.01	—	8.03	2k: job p50 239 s, max 553 s; 1 failure
EPYC 9575F · fixes off	8.13	8.02	8.04	—	—	
Graviton5 · fixes on	23.62	22.33	22.89	—	—	
Graviton5 · fixes off (full ladder, 4 KiB)	23.76	22.63	22.73	22.79	22.96	0 failures to 2,000
Turin	not run	not run	not run	not run	not run	the decisive missing row
Vera one socket (published Sep)	11.8–13.7	12.9–13.3	12.8–13.1	12.8–13.0	12.7–14.2	2k had 53–627 failures
EPYC 9755 (published)	9.6–10.0	10.7–11.5	10.7–11.3	10.7–11.3	10.9–12.0	failures from 1,408

[d] the fixes-off 176 rung read 1.47 jobs/s because of wedged guests; 15.0 is the clean rung on the same build. Level 704 on Vera and the 9575F is from a rerun with a 600 s client timeout (120 s cut off healthy 9575F jobs).

4.1. Derived: per-core and per-thread rate (plateau)

Host	Plateau jobs/s	Cores / threads	per core	per thread	Clock drop packed	Single job, light load
HQ Vera	26.7	176 / 352	**0.152**	0.076	−6 %	**5.9 s**
Graviton5	22.9	192 / 192	0.119	**0.119**	−3 %	6.65 s
EPYC 9575F	8.0	64 / 128	0.126	0.063	−24 %	6.7 s
EPYC 9755 (published)	≈ 11	128 / 256	0.086	0.043	—	7.1 s
Turin	not run	192 / 384	—	—	—	—

Platform overhead is 0.5–0.6 % of each job, so throughput is cores × per-core speed under full load. Vera's lead is per **physical core** (with SMT); per hardware thread Graviton5 is ahead. Vera's 5.9 s single-job figure is from the September one-socket data.

5. Density

Host	Result
Turin	Bursts: 3,000 in 42.7 s, 4,000 in 81.9 s, 0 failures. Waves of 2,000 kept alive: **8,000 live with 0 failures**, each wave ready in 10–15 s. Peak **9,750 live** (9,946 Firecracker processes) with 1.7 TB memory free. Limits hit on the way, all configuration: 60 s create-pickup cap (4–5k), 1 GiB scratch reservation (6,742), `RLP_MAX_QUEUED_CREATES=8000`. Final stop: the runner's heartbeat starved, the API marked the runner lost and the lost-sweep marked 9,876 live VMs terminal (evidence added to 2459). First hardware limit (CPU reservation) extrapolates to ≈ 14,600.

HQ Vera	2,000 live in the throughput ladder: 1 failure in 16,000 jobs, peak memory 470 of 1,479 GiB (≈ 170 MiB per sandbox under load). Configured cap 3,000. Beyond that not measured.
EPYC 9575F	2,000 live: 1 failure in 16,000 jobs, 393 of 1,133 GiB (≈ 184 MiB per sandbox). Configured cap 3,000; at 3,000 jobs would take ≈ 375 s each and need a longer client timeout.
Graviton5	2,000 in the ladder with 0 failures (fixes off). Memory per sandbox ≈ 129 MiB idle / 215 MiB under load on 4 KiB pages; 64 KiB pages cost ≈ 8 % more (the earlier "2.3×" was a measurement error).

6. New findings in this round

- **The fixes help x86 more than Vera.** Idle-image p50 −57 % on the 9575F and ≈ −60 % on Graviton5 vs −30 % on Vera. With the fixes on, "Vera starts sandboxes faster" is reversed. Likely cause (unmeasured): the remaining create path is serial work where higher clocks win (9575F up to 5.0 GHz vs Vera 3.36 GHz).
- **Which fix matters depends on the host.** On Turin only 3260 (init-disk fsync, 313 ⟶ 9 ms) and 3261 (cgroup pool) moved anything; 3265 targets an arm64-only 580 ms ARP stall and 3269 a listen-queue overflow that Turin never showed.
- **Turin's "8 s startup" is explained:** it was measured without the fixes. With them it is 3.9 s for 1k idle.
- **Density failure mode:** at ≈ 8–10k live sandboxes on one runner, the heartbeat starves and the lost-sweep orphans the whole fleet. Same class as 2459 (NATS-stall trigger there; host saturation here). Adoption verification is also capped at 4,096 VMs per host.
- **Graviton5:** 5 s toolbox waits in 5 of 3,000 fixes-on sandboxes (new, unexplained); pinning to one NUMA node halves throughput (cross-node memory costs 1.25× there vs 3.08× on Vera).
- **Measurement traps found and fixed:** a 120 s client read timeout reported healthy 9575F jobs as failures at 704; the SDK's `int(disk)*1024` silently reserved 1 GiB scratch per sandbox on Turin; a silently failed runner drop-in on Graviton5.

7. Gaps and caveats

Gap / caveat	Effect on this report
Turin throughput ladder not run	The deciding number for "where Vera shines". Recommended next run (≈ 1 h, fixes on, 176 ⟶ 2,000).
`RLP_VM_MAXACK` 2,048 on the 9575F vs 256 elsewhere	Runner-slot wait is Vera's largest remaining startup step (421 ms). The 9575F's startup lead may partly be this setting; rerun Vera with 2,048 (≈ 15 min).
Turin client: HTTP pool 512, 64 MiB disk	Others used 1,024 and 1 GiB. Small effect expected on startup; not equalized.
Graviton5: no 2k startup and no fixes-on ladder above 704	Graviton5′s 2k fixes-off ladder (22.96 jobs/s, 0 failures) stands in.
Page size and preemption differ	Vera 64 KiB + `preempt=none`; others 4 KiB; the AWS kernel cannot switch preemption.
Fixes are not merged	All four live only in the bench build. Fixes-on numbers must not be compared with stock EPYC data.
Vera wedged guests (3213, ≈ 0.2 %)	Inflate Vera's all-ready numbers when they hit; reported with failure counts.
EPYC 9755 unreachable	Only its published (older-build) numbers appear.

Sources: `data/vera-hq-cell-20261014/BASELINE-{VERA,EPYC9575}-FIXES-20261014.md` and `.json`; `data/phoenix-e6-cell-20261014/FINDINGS.md`, `runs.tgz`, `results-fixed.tgz` (Turin p50/p95 recomputed from the raw per-sandbox JSONL); `data/graviton5-cell-20261014/FINDINGS.md` §5–6; `daniel-focus-here/public/` for published September data.